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Display with array of light-transmitting windows

Abstract

A display may have an array of pixels. Each pixel may have a light-emitting diode such as an organic light-emitting diode. The organic light-emitting diodes may each have an anode that is coupled to a thin-film transistor pixel circuit for controlling the anode. Transparent windows may be formed in the display. The windows may be formed by replacing data storage capacitors and other pixel circuit structures in a subset of the pixels with transparent window structures, by selectively removing portions of light-emitting diode anodes, and by shifting anodes. An array of electrical components such as an array of light sensors may be aligned with the transparent windows and may be used to measure light passing through the transparent windows.

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Background/Summary

(1) This application is a continuation of patent application Ser. No. 16/859,904, filed Apr. 27, 2020, now U.S. Pat. No. 11,417,709, which is a continuation of patent application Ser. No. 15/257,374, filed Sep. 6, 2016, now U.S. Pat. No. 10,644,077, which claims the benefit of provisional patent application No. 62/247,543, filed Oct. 28, 2015, all of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

(1) This relates generally to electronic devices, and, more particularly, to electronic devices with displays.

(2) Electronic devices often include displays. Displays such as organic light-emitting diode displays have pixels with light-emitting diodes. The light emitting diodes each have electrodes (i.e., an anode and a cathode). Emissive material is interposed between the electrodes. During operation, current passes through the emissive material between the electrodes, generating light.

(3) The electrodes in an organic light-emitting diode display are formed from a photolithographically patterned layer of conductive material. Electrodes are organized in a regularly spaced array. This type of arrangement simplifies the layout of thin-film transistor circuits for the display.

(4) It may be desirable to incorporate electrical components into a display. If care is not taken, the electrodes and other circuitry in a display may interfere with these components.

(5) It would therefore be desirable to be able to provide improved display arrangements for accommodating the addition of electrical components.

SUMMARY

(6) A display may have an array of pixels. Each pixel may have a light-emitting diode such as an organic light-emitting diode. The organic light-emitting diodes may each have an anode that is coupled to a thin-film transistor pixel circuit for controlling the anode.

(7) Transparent windows may be formed in the display. The windows may be formed by replacing data storage capacitors and other pixel circuit structures in some of the pixels with transparent window structures. If desired, windows may be accommodated by selectively removing portions of the anodes in some of the pixels. Configurations may also be used in which the anodes in some of the pixels are shifted relative to the anodes in the other pixels.

(8) An array of electrical components may be aligned with the transparent windows. For example, the display may have an array of light transmitting windows each of which is aligned with a respective light detector that measures light passing through that light transmitting window.

(9) Further features will be more apparent from the accompanying drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of an illustrative electronic device having a display in accordance with an embodiment.
- (2) FIG. 2 is a top view of an illustrative display in an electronic device in accordance with an embodiment.
- (3) FIG. 3 is a cross-sectional side view of a portion of an illustrative organic light-emitting diode display in accordance with an embodiment.
- (4) FIG. 4 is a cross-sectional side view of a display with an array of electrical components in accordance with an embodiment.
- (5) FIG. 5 is a top view of an illustrative display with an array of electrical components mounted behind windows that are located within selected pixels in the display in accordance with an embodiment.
- (6) FIG. 6 is a circuit diagram of an illustrative display having modified pixels to accommodate windows for light-based components in accordance with an embodiment.
- (7) FIG. 7 is a circuit diagram of an illustrative display showing how a pixel circuit formed in an inactive display area may be used to control a light-emitting diode in a pixel in which pixel circuitry has been removed to accommodate a light transmitting window in accordance with an embodiment.
- (8) FIG. 8 is a cross-sectional side view of thin-film transistor circuitry of the type that may be associated with normal pixels on a display in accordance with an embodiment.
- (9) FIG. 9 is a cross-sectional side view of thin-film transistor circuitry of the type that may be associated with pixels in which pixel circuitry has been removed to accommodate light transmitting windows in accordance with an embodiment.
- (10) FIG. 10 is a top view of an illustrative display having an array of diamond-shaped pixels showing how pixels may be selectively modified to accommodate light transmitting windows in accordance with an embodiment.

DETAILED DESCRIPTION

- (11) An illustrative electronic device of the type that may be provided with a display is shown in FIG. 1. As shown in FIG. 1, electronic device 10 may have control circuitry 16. Control circuitry 16 may include storage and processing circuitry for supporting the operation of device 10. The storage and processing circuitry may include storage such as hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry 16 may be used to control the operation of device 10. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, application specific integrated circuits, etc.
- (12) Input-output circuitry in device 10 such as input-output devices 12 may be used to allow data to be supplied to device 10 and to allow data to be provided from device 10 to external devices. Input-output devices 12 may include buttons, joysticks, scrolling wheels, touch pads, key pads, keyboards, microphones, speakers, tone generators, vibrators, cameras, sensors, light-emitting

diodes and other status indicators, data ports, and other electrical components. A user can control the operation of device **10** by supplying commands through input-output devices **12** and may receive status information and other output from device **10** using the output resources of input-output devices **12**.

(13) Input-output devices **12** may include one or more displays such as display **14**. Display **14** may be a touch screen display that includes a touch sensor for gathering touch input from a user or display **14** may be insensitive to touch. A touch sensor for display **14** may be based on an array of capacitive touch sensor electrodes, acoustic touch sensor structures, resistive touch components, force-based touch sensor structures, a light-based touch sensor, or other suitable touch sensor arrangements.

(14) Control circuitry **16** may be used to run software on device **10** such as operating system code and applications. During operation of device **10**, the software running on control circuitry **16** may display images on display **14** using an array of pixels in display **14**.

(15) Device **10** may be a tablet computer, laptop computer, a desktop computer, a display, a cellular telephone, a media player, a wristwatch device or other wearable electronic equipment, or other suitable electronic device.

(16) Display **14** may be an organic light-emitting diode display or may be a display based on other types of display technology. Configurations in which display **14** is an organic light-emitting diode display are sometimes described herein as an example. This is, however, merely illustrative. Any suitable type of display may be used, if desired.

(17) Display **14** may have a rectangular shape (i.e., display **14** may have a rectangular footprint and a rectangular peripheral edge that runs around the rectangular footprint) or may have other suitable shapes. Display **14** may be planar or may have a curved profile.

(18) A top view of a portion of display **14** is shown in FIG. 2. As shown in FIG. 2, display **14** may have an array of pixels **22** formed on substrate **36**. Substrate **36** may be formed from glass, metal, plastic, ceramic, or other substrate materials. Pixels **22** may receive data signals over signal paths such as data lines **D** and may receive one or more control signals over control signal paths such as horizontal control lines **G** (sometimes referred to as gate lines, scan lines, emission control lines, etc.). There may be any suitable number of rows and columns of pixels **22** in display **14** (e.g., tens or more, hundreds or more, or thousands or more). Each pixel **22** may have a light-emitting diode **26** that emits light **24** under the control of a pixel circuit formed from thin-film transistor circuitry such as thin-film transistors **28** and thin-film capacitors). Thin-film transistors **28** may be polysilicon thin-film transistors, semiconducting-oxide thin-film transistors such as indium gallium zinc oxide transistors, or thin-film transistors formed from other semiconductors. Pixels **22** may contain light-emitting diodes of different colors (e.g., red, green, and blue diodes for red, green, and blue pixels, respectively) to provide display **14** with the ability to display color images.

(19) Pixels **22** may be arranged in a rectangular array or an array of other shapes. The array of pixels **22** forms an active area **AA** for display **14** and is used in displaying images for a user. Inactive portions of display **14** such as inactive area **IA** may run along one or more of the edges of active area **AA**. Inactive area **IA** may be free of pixels **22**.

(20) Display driver circuitry may be used to control the operation of pixels **22**. The display driver circuitry may be formed from integrated circuits, thin-film transistor circuits, or other suitable circuitry and may be located in inactive area **IA**. Display driver circuitry **30** of FIG. 2 may contain communications circuitry for communicating with system control circuitry such as control circuitry **16** of FIG. 1 over path **32**. Path **32** may be formed from traces on a flexible printed circuit or other cable. During operation, the control circuitry (e.g., control circuitry **16** of FIG. 1) may supply circuitry **30** with information on images to be displayed on display **14**.

(21) To display the images on display pixels **22**, display driver circuitry **30** may supply image data to data lines **D** while issuing clock signals and other control signals to supporting display driver circuitry such as gate driver circuitry **34** over path **38**. If desired, circuitry **30** may also supply clock

signals and other control signals to gate driver circuitry on an opposing edge of display **14**.

(22) Gate driver circuitry **34** (sometimes referred to as horizontal control line control circuitry) may be implemented as part of an integrated circuit and/or may be implemented using thin-film transistor circuitry. Horizontal control lines G in display **14** may carry gate line signals (scan line signals), emission enable control signals, and other horizontal control signals for controlling the pixels of each row. There may be any suitable number of horizontal control signals per row of pixels **22** (e.g., one or more, two or more, three or more, four or more, etc.).

(23) A cross-sectional side view of a portion of an illustrative organic light-emitting diode display that includes a light-emitting diode (diode **26**) and thin-film transistor circuitry **48** for an associated pixel circuit is shown in FIG. **3**. As shown in FIG. **3**, display **14** may include a substrate layer such as substrate layer **36**. Substrate **36** may be a planar layer or a non-planar layer and may be formed from plastic, glass, ceramic, sapphire, metal, or other suitable materials. The surface of substrate **36** may, if desired, be covered with one or more buffer layers (e.g., inorganic buffer layers such as layers of silicon oxide, silicon nitride, etc.).

(24) Thin-film transistor circuitry **48** may be formed on substrate **36**. The thin film transistor circuitry may include transistors, capacitors, and other thin-film structures. As shown in FIG. **3**, a transistor such as thin-film transistor **28** may be formed from thin-film semiconductor layer **60**. Semiconductor layer **60** may be a polysilicon layer, a semiconducting-oxide layer such as a layer of indium gallium zinc oxide, or other semiconductor layer. Gate layer **56** may be a conductive layer such as a metal layer that is separated from semiconductor layer **60** by an intervening layer of dielectric such as dielectric **58** (e.g., an inorganic gate insulator layer such as a layer of silicon oxide). Dielectric **62** may also be used to separate semiconductor layer **60** from underlying structures such as shield layer **64** (e.g., a shield layer that helps shield the transistor formed from semiconductor layer **60** from charge in buffer layers on substrate **36**).

(25) Semiconductor layer **60** of transistor **28** may be contacted by source and drain terminals formed from source-drain metal layer **52**. Dielectric layer **54** (e.g., an inorganic interlayer dielectric layer) may separate gate metal layer **56** from source-drain metal layer **52**. Pixel circuits formed from thin-film transistor circuitry **48** may be shorted to anode **42** of light-emitting diode **26** using a metal via such as via **53**. Via **53** may, for example, be coupled to transistor structures such as source-drain metal layer **52** and may pass through dielectric planarization layer **50**. Planarization layer **50** may be formed from an organic dielectric material such as a polymer.

(26) Light-emitting diode **26** is formed from light-emitting diode layers **40** on thin-film transistor layers **48**. Each light-emitting diode has a lower electrode and an upper electrode. In a top emission display, the lower electrode may be formed from a reflective conductive material such as patterned metal to help reflect light that is produced by the light-emitting diode in the upwards direction out of the display. The upper electrode (sometimes referred to as the counter electrode) may be formed from a transparent or semi-transparent conductive layer (e.g., a thin layer of transparent or semitransparent metal and/or a layer of indium tin oxide or other transparent conductive material). This allows the upper electrode to transmit light outwards that has been produced by emissive material in the diode. In a bottom emission display, the lower electrode may be transparent (or semi-transparent) and the upper electrode may be reflective.

(27) In configurations in which the anode is the lower electrode, layers such as a hole injection layer, hole transport layer, emissive material layer, and electron transport layer may be formed above the anode and below the upper electrode, which serves as the cathode for the diode. In inverted configurations in which the cathode is the lower electrode, layers such as an electron transport layer, emissive material layer, hole transport layer, and hole injection layer may be stacked on top of the cathode and may be covered with an upper layer that serves as the anode for the diode. Both electrodes may reflect light.

(28) In general, display **14** may use a configuration in which the anode electrode is closer to the display substrate than the cathode electrode or a configuration in which the cathode electrode is

closer to the display substrate than the anode electrode. In addition, both bottom emission and top emission arrangements may be used. Top emission display configurations in which the anode is located on the bottom and the cathode is located on the top are sometimes described herein as an example. This is, however, merely illustrative. Any suitable display arrangement may be used, if desired.

(29) In the illustrative configuration of FIG. 3, display **14** has a top emission configuration and lower electrode **42** is an anode and upper electrode **46** is a cathode. Layers **40** include a patterned metal layer that forms anodes such as anode **42**. Anode **42** is formed within an opening in pixel definition layer **66**. Pixel definition layer **66** may be formed from a patterned photoimageable polymer such as polyimide. In each light-emitting diode, organic emissive material **44** is interposed between a respective anode **42** and cathode **46**. Anodes **42** may be patterned from a layer of metal on a planarization layer in the thin-film transistor layers of pixel circuit **48** such as planarization layer **50**. Cathode **46** may be formed from a common conductive layer that is deposited on top of pixel definition layer **66**. Cathode **46** is transparent so that light **24** may exit light emitting diode **26** as current is flowing through emissive material **44** between anode **42** and cathode **46**.

(30) Display **14** may have an array of pixels **22** of different colors to provide display **14** with the ability to display color images. The pixels may include red pixels, green pixels, and blue pixels. White pixels, yellow pixels, and pixels of other colors may also be included in display **14**, if desired. The pixels may have rectangular emissive areas (e.g., rectangular anode shapes) and/or may have emissive areas of other suitable shapes (e.g., diamond shapes, etc.).

(31) It may be desirable to incorporate electrical components into display **14** and/or device **10**. As shown in FIG. 4, for example, electrical components **84** may be incorporated into device **10** under pixels **22**. Components **84** may be discrete components or may be formed as part of a common integrated circuit or other shared component (see, e.g., component **82**, which may be an integrated circuit). If desired, components **84** may be mounted on a common substrate such as a printed circuit (e.g., a component **82** may be a rigid printed circuit board formed from a rigid printed circuit board material such as fiberglass-filled epoxy or a flexible printed circuit formed from a flexible layer of polyimide or other sheet of polymer). Components **84** may be integrated into the layers that make up display **14** and/or may be mounted in alignment with display **14**.

(32) Electrical components **84** may be audio components (e.g., microphones, speakers, etc.), radio-frequency components, haptic components (e.g., piezoelectric structures, vibrators, etc.), may be capacitive touch sensor components or other touch sensor structures, may be temperature sensors, pressure sensors, magnetic sensors, or other sensors, or may be any other suitable type of electrical component. With one suitable arrangement, which may sometimes be described herein as an example, electrical components **84** may be light-based components (e.g., components that emit and/or detect visible light, infrared light, and/or ultraviolet light).

(33) Light-based components **84** may emit and/or detect light that passes through transparent windows **76** in display **14**. Windows **76** may be formed between the pixel circuitry and emissive areas of pixels **22** and may include transparent materials (e.g., clear plastic, glass, etc.) and/or holes (e.g., air-filled openings or openings filled with transparent material that pass partly or fully through substrate **36** and other display layers **74** of display **14**). There may be a window **76** between each set of adjacent pixels **22** or, more preferably, blocks of pixels **22** (e.g., blocks of tens, hundreds, or thousands of pixels) may be associated with windows **76** and electrical components **84**. For example, additional space may be created within some of pixels **22** to accommodate windows **76** and components **84**. There may be, as an example, a window **76** associated with each block of 100 pixels **22** or other suitable number of pixels **22** (e.g., each 50 pixels or more, each 200 pixels or more, each 100 pixels or more, each 50 or fewer pixels, each 200 or fewer pixels, etc. If desired, some components may be mounted on the upper surface of display **14** (in which cases windows **76** need not be provided through layers **74** of display **14**).

(34) Examples of light-based components **84** that emit light include light-emitting diodes (e.g.,

organic light-emitting diodes, discrete crystalline light-emitting diode dies, etc.), lasers, and lamps. Examples of light-based components that detect light include light detectors such as photodiodes and phototransistors. Some components may, if desired, include both light emitters and detectors. For example, components **84** may emit infrared light and may include light detector structures for detecting a portion of the emitted light that has reflected from nearby objects such as object **86**. Components of this type may be used to implement a proximity detector, a light-based fingerprint sensor (e.g., when object **86** is the finger of a user), or other light-based sensor. If desired, light-based sensors such as these may be implemented by illuminating object **86** with light **24** from one or more of pixels **22** and/or light **78** from one or more supplemental light sources such as discrete light-emitting diodes **80**, while using light-detecting components **84** to gather reflected light from object **86**.

(35) Control circuitry **16** may be used in controlling the emission of light from light sources such as pixels **22**, components **84**, and/or light sources **80** and may be used in processing corresponding detected light from components **84** (e.g., to generate a proximity sensor signal based on light reflected from object **86**, to generate a fingerprint reading based on light reflected from object **86**, to process a captured digital image of a far-field object, that is captured using components **84**, etc.).

(36) Components **84** (and windows **76**, if used) may be interspersed with pixels **22** using any suitable arrangement. With one illustrative configuration, which is shown in FIG. 5, windows **76** and components **84** are arranged in an array that has a larger pitch than the array of pixels **22** in display **14**. There may be, for example, one window **76** and one corresponding component **84** for each set of 10-1000 pixels **22**, for each set of 100-10,000 pixels, for each set of more than 500 pixels, or for each set of less than 5000 pixels (as examples). In configurations such as these, pixels **22** are arranged on display **14** with a finer pitch than windows **76** and components **84**. Pixels **22** may, for example, be organized in an array having rows and columns and windows **76** and components **84** may be arranged in an array having a smaller number of rows and columns. Configurations in which windows **76** and components **84** are arranged in patterns other than rectangular arrays may also be used. Arrangements in which windows **76** and components **84** are arranged in rows and columns may sometimes be described herein as an example. Windows **76** may cover all of active area AA or may be confined to a portion of active area AA. If desired, some of windows **76** may extend into inactive area IA.

(37) To make room for windows **76** in the pixels of active area AA, it may be desirable to remove some of the normal pixel circuitry of some of these pixels. For example, pixels **22A** of FIG. 5 may contain complete pixel circuits for controlling corresponding light-emitting diodes **26** (i.e., pixel circuits that contain a full complement of thin-film transistors and capacitors), whereas pixels **22B** may contain incomplete pixel circuits (i.e., circuits similar to the circuits of pixels **22A** from which one or more transistors and/or capacitors or all of the transistors and capacitors of pixels **22A** have been removed and replaced with windows **70**). By removing some or all of the normal pixel circuitry used in controlling the light-emitting diode of each of pixels **22B**, additional area may be made available in pixels **22B** to accommodate windows **76**. To make up for the removed pixel circuitry of pixels **22B**, ancillary pixel circuits can be formed in inactive area IA along the edge of display **14**. Signal lines may then be used to route signals between the light-emitting diodes of pixels **22B** and the ancillary pixel circuits in the inactive area.

(38) FIG. 6 is a diagram of an illustrative display in which pixel circuitry has been removed from pixel **22B** in active area AA and replaced with a window **76**. The functions of the removed pixel circuitry are performed by ancillary pixel circuitry **102** in inactive area IA.

(39) As shown in FIG. 6, each normal pixel **22A** has a light-emitting diode **24** and an associated pixel circuit **100**. The pixel circuit **100** of each normal pixel **22A** includes a drive transistor that is coupled in series with the light-emitting diode **24** of that pixel for controlling the amount of drive current that passes through that light-emitting diode. Switching transistors and one or more capacitors (e.g., data storage capacitors for storing loaded data signals, etc.) may also be included

in each pixel circuit **100**.

(40) In windowed pixels such as pixel **22B** of FIG. **6**, some or all of the thin-film circuitry associated with circuit **100** is removed, thereby creating a transparent area through display **14** for window **76**. During operation of display **14**, data signals for pixel **22B** may be routed to ancillary pixel circuit **102** using ancillary data line **104**. Pixel circuitry **102** may have an ancillary drive transistor and/or other ancillary pixel circuitry (e.g., an ancillary data storage capacitor) that uses the data signal on ancillary data line **104** to produce a corresponding drive current I_d for light-emitting diode **24** of pixel **22B**. Ancillary drive current path (line) **106** may route drive current I_d to light-emitting diode **24** in pixel **22B** from ancillary pixel circuit **102**.

(41) Other techniques may be used for making up for the pixel circuitry removed from pixel **22** under window **76** if desired. The use of ancillary pixel circuitry such as circuitry **102** in inactive area **IA** and supplemental signal lines such as ancillary data line **104** and ancillary drive current line **106** is merely illustrative. Moreover, other modifications may be made to pixels **22B** to help accommodate windows **76** in addition to or instead of replacing pixel circuitry with windows **76** such as selectively removing portions of anodes **42** and/or shifting anodes **42** in pixels **22B** relative to the anodes **42** in normal pixels **22A**, etc.

(42) An example of circuitry that may be used for display **14** is shown in FIG. **7**. In the FIG. **7** example, each pixel circuit may contain a drive transistor **TD**, a data storage capacitor **Cst**, and a switching transistor **SW**. Other types of pixel circuits may be used in pixels **22**, if desired. For example, the pixel circuits of pixels **22A** and ancillary circuits **102** may have additional switching transistors, emission control transistors, additional capacitors, etc. The pixel circuitry of FIG. **7** is merely illustrative.

(43) In normal pixels **22A** of FIG. **7**, a full set of pixel circuit components is present (e.g., drive transistor **TD**, switching transistor **SW**, and storage capacitor **Cst**). In pixels **22B**, some or all of these pixel circuit components may be removed and replaced with window **76**. To make up for the pixel circuit components removed from pixel **22B**, ancillary pixel circuit **102** may be formed in inactive area **IA** that contains the removed components. For example, in a scenario in which switching transistor **SW**, drive transistor **TD**, and storage capacitor **Cst** have been removed from pixel **22B**, ancillary pixel circuit **102** may include switching transistor **SW**, drive transistor **TD**, and storage capacitor **Cst**.

(44) During operation, gate line **G** may be asserted, thereby turning on switching transistor **SW** in ancillary pixel circuit **102** and loading data from data line **D** onto capacitor **Cst** via ancillary data line **104** and switching transistor **SW**. This establishes a desired voltage V_g on the gate of drive transistor **TD** in circuit **102** and thereby establishes a desired value for drive current I_d . Ancillary pixel circuit **102** is located in inactive area **IA** and pixel **22B** is located in active area **AA**. To ensure that light-emitting diode **24** in pixel **22B** emits a desired amount of light, ancillary drive current line **106** may be used to route drive current I_d to light-emitting diode **24** in pixel **22B** from circuit **102**.

(45) FIG. **8** is a cross-sectional side view of illustrative thin-film transistor circuitry of the type that may be used in normal pixels **22A**. As shown in FIG. **8**, pixel circuitry for pixel **22A** may include transistors such as transistor **28**, an anode such as anode **42** for light-emitting diode **24**, and structures for forming storage capacitor **Cst**. In the example of FIG. **8**, storage capacitor **Cst** has a first electrode (electrode **56'**) that is formed from a portion of the gate metal layer used in forming transistor gate **56** in transistor **28** and a second electrode (electrode **110**) formed from a second gate metal layer that is interposed between interlayer dielectric layer **54-1** and interlayer dielectric layer **54-2** of layer **54**. Other storage capacitor arrangements may be used, if desired. Pixel **22A** may be free of transparent window structures.

(46) To make room for window **76**, transistor structures and other structures may be removed from some of the pixels in display **14** as described in connection with pixel **22B** of FIG. **7** (i.e., pixels **22B** may be free of data storage capacitors and may, if desired, be free of switching transistors

and/or drive transistors). FIG. 9 is a cross-sectional side view of illustrative thin-film transistor circuitry of the type that may be used in pixels 22B that contain windows 76. As shown in FIG. 9, the electrode structures associated with capacitor Cst may be removed, so that region 76 is transparent and can serve as light-transparent window 76 in place of capacitor Cst. In the area normally used for forming transistors, layers such a layer 60 may be patterned to form signal interconnects to route other signals between components in remaining circuitry in pixel 22). Supplemental paths 106 (e.g., supplemental drive current lines such as supplemental lines 106 of FIGS. 6 and 7) may be formed from a patterned portion of the gate metal layer (layer 56 of normal pixel circuits 22A) or may, as illustrated by optional path 106' be formed from a portion of the second gate metal layer (i.e., the layer used to form electrode 110 of FIG. 8). The second gate metal layer may be interposed between interlayer dielectric layers 54-1 and 54-2.

(47) Pixels 22 may be arranged in an RGB stripe pattern or other suitable pattern. Pixels 22 may include pixels such as red pixels, green pixels, and blue pixels each of which may have anodes of the same size or different sizes (e.g., the red and green anodes may be different in size from the blue anodes). If desired, windows 76 may be formed in displays with diamond shaped emissive areas, as illustrated by the diamond-shaped pixels of FIG. 10. In the example of FIG. 10, display 14 has an array of diamond shaped emissive areas in red pixels R, green pixels G, and blue pixels B.

(48) By deleting some or all of the components of pixel 22B (e.g., storage capacitor Cst and, if desired, one or more transistors in the pixel circuitry of pixel 22B) from window regions such as illustrative window region 76 in pixel 22B, an array of windows 76 may be formed in display 14. Windows such as window 76 may be formed in blue pixels or in pixels of other colors (e.g., red pixels or green pixels). If desired, a portion of anode 42 such as portion 42' may be removed to enlarge window 76 (i.e., the anodes 42 in pixels 22B may be smaller than the anodes of the same color in pixels 22A). Window 76 can also be accommodated and/or enlarged by shifting the location of anode 42 relative to the anodes in normal pixels 22A, as indicated by arrow 120 and alternate anode location 42A. Combinations of these approaches (e.g., deletion of one or more pixel circuit components from area 76 adjacent to anode 42, enlargement of area 76 by removing portion 42' of anode 42, and/or shifting anode 42 to location 42A) may be used to accommodate windows in display 14, if desired. In displays with pixels of different colors (e.g., red, green, blue, etc.), anodes 42 for one or more (or all) of the different colored pixels may be shifted, may be reduced in size, and/or may have associated deleted pixel circuitry, anodes in adjacent pixels may be shifted away from each other to help make space for window 76, etc.

(49) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device, comprising: a housing; a display mounted in the housing and having an array of pixels including first and second pixels, wherein the second pixel has at least one pixel circuit element that is excluded from the first pixel to accommodate a light-transmissive window in the display; and a light sensor located behind the array of pixels, wherein the light sensor detects light through the light-transmissive window in the display.
2. The electronic device defined in claim 1 wherein the pixels comprise organic light-emitting diode pixels.
3. The electronic device defined in claim 1 wherein the display comprises a substrate on which the array of pixels is located.
4. The electronic device defined in claim 3 wherein the light-transmissive window comprises a hole in the substrate.
5. The electronic device defined in claim 4 wherein the hole passes from an upper surface to a

lower surface of the substrate.

6. The electronic device defined in claim 1 wherein the display has an active area and an inactive area, wherein the array of pixels is located in the active area, and wherein the first pixel has pixel circuitry that is located in the inactive area of the display.

7. The electronic device defined in claim 6 wherein the pixel circuitry associated with the first pixel comprises a drive current line that extends between the inactive area and the first pixel.

8. The electronic device defined in claim 7 wherein the second pixel includes a data storage capacitor and wherein the first pixel is free of a data storage capacitor.

9. The electronic device defined in claim 3 wherein the light-transmissive window is part of an array of light-transmitting windows in the substrate.

10. The electronic device defined in claim 1 wherein the light sensor is part of an array of light sensors that respectively detect the light through the array of light-transmitting windows.

11. An electronic device, comprising: a display comprising an array of pixels on a substrate; light-transmitting windows within the array of pixels, wherein the array of pixels comprises first and second groups of pixels and wherein the second group of pixels has pixel circuit elements that are excluded from the first group of pixels to accommodate the light-transmitting windows; and a light sensor behind the substrate that detects light that passes through the light-transmitting windows within the array of pixels.

12. The electronic device defined in claim 11 wherein the first group of the pixels has less pixel circuitry than the second group of the pixels.

13. The electronic device defined in claim 12 wherein the display comprises an active area and an inactive area and wherein at least some of the pixel circuitry associated with the first group of the pixels is located in the inactive area of the display.

14. The electronic device defined in claim 13 wherein the first group of pixels do not include storage capacitors.

15. The electronic device defined in claim 11 wherein the light-transmitting windows comprise holes in the substrate.

16. An electronic device, comprising: a housing; a display mounted in the housing, wherein the display has an active area with an array of pixels including first and second groups of pixels, wherein the second group of pixels has pixel circuit elements that are excluded from the first group of pixels; and an array of light sensors located behind the first group of pixels, wherein the array of light sensors receives light that passes through the display.

17. The electronic device defined in claim 16 wherein the array of light sensors forms a fingerprint sensor.

18. The electronic device defined in claim 16 wherein the display comprises light-transmissive windows within the array of pixels through which the light passes.

19. The electronic device defined in claim 16 wherein the display has an inactive area without pixels and wherein the first group of the pixels has at least some pixel circuitry that is located in the inactive area.

20. The electronic device defined in claim 16 wherein the pixels in the first group do not include storage capacitors.
